MILITARY SPECIFICATION

AUGER MACHINES, PIPE AND SEWER CLEANER: GASOLINE-ENGINE-DRIVEN, TRAILER MOUNTED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Classification. The auger machines shall be of the following sizes as specified (see 6.2):

Size 500 - Minimum capacity of 500 feet.
Size 800 - minimum capacity of 800 feet.
Size 1000 - Minimum capacity of 1,000 feet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-281 - Automobile, Trucks, Truck-Tractors and Trailer Dollies, Preservation and Packaging of.
MS51336 - Lunette and Coupler, Drawbar, Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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2.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design. The auger machine shall include all features, components, and accessories necessary for opening clogged sewer and drain pipes. The auger machine shall include a reel assembly, rod drive mechanism, rods, gasoline-driven-engine, and hydraulic system, all mounted on a two-wheeled trailer.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete auger for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The auger machine shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features

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which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the auger machine being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Reel assembly. The reel assembly shall be mounted within a balanced, bearing supported frame which rotates about the axis of rod travel direction.

3.7 Rod drive mechanism. The auger machine shall have a rod drive mechanism with a positive drive dog mechanism which engages each coupling of the sectional sewer rods and provides equal driving or pulling action upon the rod couplings. The drive mechanism on the size 800 and 1000 units shall have the capability of automatically censoring and obstruction, at which point it withdraws the rods and auger at a pre-set speed and distance. The unit shall have the capability of continuing this function until the obstruction is eliminated. The size 800 foot and 1000 foot auger machines drive mechanism shall accommodate rods 48 inches in length. The drive mechanism for size 500 shall accommodate rods 39 inches in length.

3.8 Rods. The rods shall be constructed of tempered steel and shall be 5/16 inch in diameter by 39 inches in length. The rods shall couple together to form a continuous length. The size 500 auger machine shall be furnished with 500 feet of rods coupled and loaded in th reel assembly. The size 800 auger machine shall be furnished with 800 feet of rods coupled and loaded in the reel assembly. The size 1000 auger machine shall be equipped with 1,000 feet of rods coupled and loaded in the reel assembly.

3.9 Engine. The auger machine shall be driven by a gasoline engine of the size and type used or recommended for use by the auger machine manufacturer. The engine shall have an intermittent horsepower (hp) rating sufficient to properly and adequately operate the auger machine and all connected auxiliaries. The engine shall be provided with the manufacturer's standard

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fuel tank and fuel filter system. An ignition switch, choke, throttle, and ammeter shall be provided and shall be mounted on a panel or panels located near the operating controls and visible to the operator. The engine shall be equipped with an electric starting system. The minimum hp rating of the engine furnished for size 500 shall be 8 hp; for size 800, 18 hp; and for size 1000, 20 hp.

3.10 Hydraulic system. The auger machine shall be fully hydraulic and shall transfer the engine hp through a hydraulic system with a minimum reservoir capacity of 5.4 gallons, baffled and vented to provide forced circulation for maximum cooling and a full flow maximum 10 micron filter installed in the hydraulic circuit. Size 500 and size 800 auger machines shall have a hydraulic gear pump with a minimum rating of 800 gallons per minute (gpm). Size 1000 auger machine shall have dual hydraulic gear pumps with a minimum rating of 15 gpm for each pump. Hydraulic motors, one each for the reel assembly and the rod drive mechanism, shall be furnished and shall have adequate ratings to assure proper operating for the engine hp and pump output. The hydraulic motors shall be capable of operating simultaneously or independently.

3.11 Controls. The auger machine shall be furnished with the controls specified herein as a minimum. A control panel shall be provided for mounting operating controls. All operating controls shall be identified by permanent, legible markings.

3.11.1 Rotation. The auger machine shall be equipped with a variable speed, hydraulically controlled circuit which will rotate the rods and the reel assembly from zero to not less than 150 revolutions per minute during various operating conditions. The auger machine shall have a control to actuate the rotation in clockwise or counterclockwise direction.

3.11.2 Safety brake. The auger machine shall be equipped with a hydraulic system which provides braking action against rotation of the rods and reel assembly.

3.11.3 Rod travel. The auger machine shall be equipped with a control to actuate the push and pull of the sewer rods. The auger machine shall be furnished with an adjustable full-time hydraulic pressure relief valve for controlling the rod travel pressure limit. An automatic footage counter circuit shall also be provided, with the capability of stopping the rod travel at a pre-determined distance.

3.12 Accessories. The auger machine shall be furnished with the minimum accessories specified in table I, as applicable.

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TABLE I. Accessories.

Item No.	Quantity (each)	Description	Furnished on Size 500	Size 800	Size 1,000
1	1	Hose rod guide, 7-foot	X	X	X
2	1	Hose rod guide, 18-foot	X	X	X
3	1	Rod guide bell	X	X	X
4	1	Rod guide brace	X	X	X
5	1	Auger, 3-inch	X	X	X
6	1	Auger, 4-inch	X	X	X
7	1	Auger, 6-inch	X	X	X
8	1	Auger, 8-inch		X	X
9	1	Square bar corkscrew, 4-inch	X	X	X
10	1	Square bar corkscrew, 6-inch	X	X	X
11	1	Square bar corkscrew, 8-inch	X	X	X
12	1	Square bar corkscrew, 10-inch			X
13	1	Cutter, minimum 3 blades, 4-inch	X	X	X
14	1	Cutter, minimum 3 blades, 6-inch	X	X	X
15	1	Cutter, minimum 3 blades, 8-inch		X	X
16	1	Cutter, minimum 3 blades, 10-inch			X
17	1	Heavy duty root saw, 4-inch	X	X	X
18	1	Heavy duty root saw, 6-inch	X	X	X
19	1	Heavy duty root saw, 8-inch		X	X
20	1	Heavy duty root saw, 10-inch			X
21	2	5/16-inch assembly tools	X	X	X
22	2	5/16-inch assembly handles	X	X	X
23	1	5/16-inch rod pick up tools	X	X	X

3.13 Trailer. The auger machine shall be mounted on a two-wheeled cart-type trailer chassis.

3.13.1 Performance. Trailer performance shall be demonstrated when fully equipped with all attachments and accessories, uniformly distributed over trailer bed load area, with complete servicing, when towed over improved and unimproved hard uneven terrain. Trailer shall show no evidence of failure or malfunction of trailer and trailer components, attachments, and accessories.

3.13.1.1 Travel speed. The trailer shall be capable of being towed over unimproved roads and reasonably hard, uneven terrain at a speed of 10 miles per hour (mph), and shall be capable of being towed over improved roads at a speed of 50 mph.

3.13.1.2 Turning ability. When towed, the trailer shall be capable of forming an angle of 50 degrees (°) to the towing vehicle, without cramping or interference to the trailer or towing vehicle.

3.13.1.3 Tracking ability. The trailer shall not exceed the tracking allowance specified in the DoT Federal Motor Carrier Safety Regulations.

3.13.2 Frame. The chassis frame shall be the manufacturer's standard for the equipment furnished.

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3.13.3 Suspension. The trailer shall be equipped with a suspension system having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the trailer loaded as specified in 3.13.1. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted.

3.13.4 Axle. Axle rating shall be at least equal to the load imposed on the axle, measured at the ground with the trailer loaded as specified in 3.13.1.

3.13.5 Wheels, rims, tires, and tubes. The trailer shall be equipped with single wheels. Rim and tire ratings shall conform to TRA recommendations for the type and size tires furnished. Tire and rim sizes shall be the same for all wheels of each trailer.

3.13.5.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of the rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground with the trailer loaded as specified in 3.13.1. Tires shall be of not less than 100 level quality.

3.13.5.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

3.13.6 Landing leg. The trailer shall have a retractable landing leg with caster type landing wheel, with cranking handle, rigid or swivel type, permanently installed. The landing leg and related parts shall withstand, without permanent deformation, the combined static and dynamic forces due to the impact during coupling and uncoupling operations. Landing leg shall operate freely, and shall be protected from malfunctioning due to foreign matter. When placed in travel position, the landing leg shall remain positively locked to provide ground clearance in the course of normal towing operations.

3.13.7 Leveling jacks. The manufacturer's standard leveling jacks, as furnished commercially, shall be provided.

3.13.8 Coupling device and towbar. The trailer shall be provided with a 3-inch lunette ring conforming to MS51336. Towbar, coupling device, and safety chains shall be provided and shall be in accordance with DoT Federal Motor Carrier Safety Regulations.

3.13.9 Fenders. A fender shall be furnished over each wheel of the trailer.

3.13.10 Toolbox. The auger shall be provided with a metal toolbox. The toolbox shall have a hinged lid which shall open not less than 90° and a latching device shall be provided. Toolbox shall be securely fastened in a protected and accessible location. Toolbox shall be capable of storing tools as specified in table I.

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3.14 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.15 Instruction plates. The auger machine shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.16 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.17 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The primer shall be applied to a clean, dry surface as soon as practible after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one auger machine when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.2 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each auger machine shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall receive the tests of 4.4.1 and 4.4.2. Each production unit shall receive the test of 4.4.2. Failure to pass any test shall constitute cause for rejection.

4.4.1 Road test. The auger machine, equipped and loaded as specified in 3.13.1, shall be road tested for a distance of not less than 10 miles to verify conformance to 3.13.1.1. During the road test, the turning ability and tracking ability shall be verified to determine conformance to 3.13.1.2 and 3.13.1.3.

4.4.2 Operational test. An operational test shall be conducted with the auger machine. The operational test shall demonstrate satisfactory performance of the engine, hydraulic system, and controls.

4.5 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packaged, and marked in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

6. NOTES

6.1 Intended use. The trailer mounted auger machines covered by this specification are intended for general maintenance use for cleaning pipe and sewers at military installations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of auger machine required (see 1.1).
c. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
d. When lubrication shall be other than specified (see 3.14).
e. Color of finish coat required (see 3.17).
f. When equipment shall be preserved, packaged, and marked in accordance with MIL-STD-281 with the level of preservation-packaging required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one auger machine. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28672B(YD)	AUGER MACHINES, PIPE AND SEWER CLEANER: GASOLINE-ENGINE-DRIVEN, TRAILER MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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